

When signals travel via ground wave, these tend to follow the earth's curvature. Medium-wave band uses this form of propagation.

3. Ionospheric propagation: Radio signal propagation through ionosphere is used by stations on short-wave bands for their signals to be heard around the globe.

4. Tropospheric propagation: In this form of radio propagation, RF signals are transmitted in the troposphere just above the earth's surface.

Besides, there are many short-range RF communication or wireless systems having radio propagation scenarios that do not fit neatly into these categories. Mobile communication systems like Wi-Fi (Wireless Fidelity), cellular systems and Bluetooth all use wireless transmission using frequencies in the radio frequency spectrum but the purpose and technology used are different.

Q2. What is piezoelectric effect? What are its applications?

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A2. Piezoelectricity (also called the piezoelectric effect) is the appearance of an electrical potential across the sides of a crystal when it is subjected to mechanical stress. The word originates from the Greek word 'piezein,' which means 'to press.'

The piezoelectric effect was discovered in 1880 by the Jacques and Pierre Curie brothers. They found that when a mechanical stress was applied on crystals like quartz, electrical charges appeared, and this voltage was proportional to the stress. Conversely, when an electric field is applied to one of the faces of the crystal, it undergoes mechanical distortion.

The effect is explained by the displacement of ions in crystals that have a non-symmetrical unit cell structure. When the crystal is compressed, the ions in each unit cell are displaced, causing electric polarisation of the unit cell. Because of the regularity of the crystalline structure, these effects

accumulate, causing the appearance of an electric potential difference between certain faces of the crystal (Fig. 2).

When an external electric field is applied to the crystal, the ions in each unit cell are displaced by electrostatic forces, resulting in mechanical deformation of the whole crystal.

The piezoelectric effect occurs only in non-conductive materials. Piezoelectric materials can be divided into two main groups: Crystals and ceramics. The most well-known piezoelectric material is quartz (SiO_2). Other examples are barium titanate and tourmaline.

There are many kinds of applications that require conversion of mechanical energy (pressure or movement of some kind) into electrical signals or vice-versa. Often, this can be done by transducers based on piezoelectricity. In ultrasound equipment, a piezoelectric transducer converts electrical energy into extremely rapid mechanical vibrations. These ultrasound vibrations can be used for scanning, cleaning, etc.

In a microphone, sound energy is converted into electrical energy with the help of piezoelectric crystals. Pressure waves from the voice make the crystal move back and forth, generating corresponding electrical signals.

In a quartz clock or watch, the reverse-piezoelectric effect is used to keep time very precisely. Electrical energy from a battery is fed into a crystal to make it oscillate thousands of times a second. The watch then uses an electronic circuit to turn that into once-per-second beats that are used by a tiny geared motor to drive the second, minute and hour hands around the clock face.

In a car's airbag sensor, a piezoelectric material detects the intensity of the shock and sends an electrical signal to trigger the airbag.

In electric cigarette lighters and spark lighters for gas stoves, squeezing of a piezoelectric crystal generates

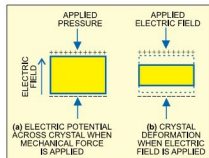


Fig. 2: Piezoelectric effect

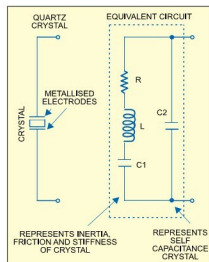


Fig. 3: Crystal equivalent circuit

voltage, making a spark fly across a small gap.

Scanning tunneling microscopes use piezoelectric crystals to scan the surface of a material and create images in great detail.

Crystal oscillators too are based on piezoelectricity. A mechanically vibrating quartz crystal can be represented by an equivalent electrical circuit consisting of low resistance, large inductance and small capacitance as shown in Fig. 3. A quartz crystal has a resonant frequency similar to that of an electrically tuned tank circuit but with a much higher Q factor.

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